

Nikah Platform

Sharia-Compliant Muslim Matrimony Platform Software Architecture Design Document

Document version	2.1 (Sprint 1 + Sprint 3 — reviewed & corrected)
Source materials	FORM_SPEC.md v1.4, Final_Architecture_design.docx (Sprints 1 & 3 draft), UMLDiagrams
Architecture style	Modular Monolith (Next.js + PostgreSQL), microservice-ready module boundaries
Reviewed by	Architecture mentorship session

This document supersedes the draft *Final_Architecture_design.docx*. Every entity, attribute, relationship and business rule below was walked through and confirmed in a structured architecture review; corrections and rationale for each change are called out inline, and unresolved items are isolated in the Open Issues appendix rather than silently decided.

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1. Revision Summary — What Changed From the Draft

The original draft (*Final_Architecture_design.docx*) and its accompanying ER diagram disagreed with each other in several places, and both were incomplete relative to the field-level requirements captured in *FORM_SPEC.md* during Sprint 1. The table below is the audit trail of every correction made and the decision that resolved it.

#	Issue found in draft	Resolution
1	Entity #9 was headed "MatchCandidte" but its attributes/relationships were actually the Guardian entity. MatchCandidate had no real definition anywhere.	Guardian restored as its own entity. MatchCandidate rebuilt from the ER diagram fields, extended with CompatibilityScore, TrustScoreAtMatch, BlendedScore, and MatchedProfileVerified.
2	PartnerPreference primary key typo: "preferpartanerId".	Renamed to PreferenceId, consistent with the rest of the schema's naming convention.
3	Admin and Report referenced in the Relationship Matrix but never defined anywhere, including the ER diagram.	No separate Admin entity created. Added Role (Candidate / Admin / Moderator) to User. Report defined as its own entity, targeting either a Profile or a Message.
4	Nine entities visible in the ER diagram (Session, OTP, LoginLog, SubscriptionPlan, Education, Occupation, Address, PrivacySetting, Attachment) had no Entity Catalog record, no business rules, no security classification.	All nine fully catalogued below with attributes, business rules and security classification.
5	Profile modeled only ~19 generic fields; none of the ~90 fields catalogued from FORM_SPEC.md in Sprint 1 (sub-sect, madhhab, hijab/beard, disability, children detail, Wali display fields, religiosity, hobbies, wedding preferences, Nafaqah/Mahr, etc.) were represented.	Profile split into six focused entities (Profile core, ReligiousPreference, MarriageIntention, ProfileChildrenDetail, plus multi-value child tables) so the schema traces cleanly back to every field in the approved form spec.
6	Two distinct "Wali" concepts — (a) someone registering on a candidate's behalf, (b) a candidate simply listing a relative's contact details for display — were conflated into one Guardian record requiring a login.	Guardian.UserId made nullable. Added TrustLevel (Registered vs. DisplayOnly) so a display-only Wali with no account is explicitly marked lower-trust than a Wali who has actually registered and verified.
7	Conversation carried a flat GuardianApprovalStatus column with no way to represent a Wali revoking, then later re-approving, a conversation — and no way to know if one or both families' guardians must sign off.	New GuardianApproval entity: one append-only row per approve/revoke decision, tagged with Side (Initiator/Receiver). Both sides' guardians gate independently — a conversation needs both green. Conversation keeps a denormalized CurrentApprovalStatus for fast checks.
8	MatchCandidate.Score conflated compatibility matching with trust/verification in an undocumented way.	CompatibilityScore (pure PartnerPreference match) and TrustScoreAtMatch are stored separately, then combined into BlendedScore. Verification-status changes update the existing row rather than requiring a new match to be generated; unverified profiles still surface in results, just flagged MatchedProfileVerified = false.
9	Height stored as two separately-updatable fields (feet and cm) in the draft's implied schema, risking the two falling out of sync.	Single canonical HeightCm column; feet/inches are a display-layer conversion only, never persisted.
10	Message carried its own ApprovalStatus column duplicating Conversation-level guardian approval, with no distinction from AI content moderation.	Message.ApprovalStatus removed. Guardian approval lives only at the Conversation/GuardianApproval level. A separate ModerationFlag on Message covers automated content moderation, a genuinely different concern.

#	Issue found in draft	Resolution
11	Draft listed Parent as a fourth primary-user actor, separate from Wali, with overlapping responsibilities.	Parent removed as a separate actor. A parent is simply one possible value of Guardian.Relation (alongside uncle, brother, etc.) — no separate entity or actor needed; Wali's interaction list now folds in "participate in communication."
12	Open Issue: Address entity's purpose overlapped ambiguously with Profile's own Country/State/City fields.	Resolved: Address now explicitly holds exactly two typed rows per profile — AddressType = Native (primary) and Current (secondary); the two may hold identical values if the person hasn't relocated. Profile.NativePlace (a duplicate free-text field) is removed; Profile.Country/State/City is retained only as a denormalized copy of the Current address, kept for fast search filtering without a join.
13	Open Issue: Admin permission granularity undecided.	Resolved for v1: Role = Admin grants full administrative permission across all modules. No separate fine-grained permission table in this version; revisit only if a real need for partial-admin roles emerges.
14	Open Issue: Photo.StorageKey encryption-at-rest undecided.	Resolved: no at-rest encryption required. Access control is enforced entirely through signed URLs and the Media module's moderation gate.

2. System Context

2.1 Actors

Actor	Type	Key Interactions
Bride / Groom	Primary user	Register, complete profile, save draft, upload photos, search/view matches, express interest, manage privacy, receive notifications
Wali (Guardian)	Primary user	Register or be listed by a candidate as any relative — father, mother, uncle, brother, etc. (Relation is free text on Guardian; Parent is one possible value, not a separate actor). Manage bride/groom profile, view suggested matches, approve or revoke conversations, participate in communication where permitted.
Admin	Internal user (User.Role)	Verify profiles, manage users & subscription plans, review reports, audit system activity
Moderator	Internal user (User.Role)	Review reported profiles, moderate messages, escalate serious cases to Admin
Payment Gateway	External system	Process payments, return payment status
SMS/Email Provider	External system	Send OTP, email & SMS notifications
Identity Verification Provider (future)	External system	Verify identity documents, return verification status
Object Storage (S3-compatible)	External system	Persist photos and message attachments; access only via signed URLs

3. Business Modules

Module boundaries are drawn so each can later be extracted into its own microservice with minimal rework — each owns a clear attribute set and a clear set of entities (Sprint 2 territory in full, summarized here for traceability).

Module	Responsibilities	Owned Entities
Authentication	Register, login, logout, password reset, OTP verification, session management	User, Session, OTP, LoginLog
User Profile	Create/edit profile, religious details, marriage intentions, background details, profile completion scoring	Profile, ReligiousPreference, MarriageIntention, ProfileChildrenDetail, ProfileDisabilityType, ProfileLanguage, ProfileHobby, Education, Occupation, Address
Partner Preference	Capture and expose what a candidate is looking for in a partner	PartnerPreference
Guardian	Wali registration, display-only Wali details, per-conversation approval/revocation	Guardian, GuardianApproval
Verification	Phone/identity/photo verification, trust scoring	VerificationRequest
Matching	Generate candidates, blend compatibility with trust, ranking	MatchCandidate
Search	Search profiles, filter, sort, paginate (reads Profile + PartnerPreference)	— (query-only over Profile/PartnerPreference)
Messaging	Conversations, messages, attachments, guardian approval gate, moderation flag	Conversation, Message, Attachment
Subscription	Plans, payments, renewals, premium access, revert/divorce/age/widowed discount rules	SubscriptionPlan, Subscription, Payment
Media	Upload/delete photos, moderation, signed URL generation	Photo
Privacy	Photo/contact/profile visibility rules	PrivacySetting
Notification	SMS, email, push, match & verification alerts	— (event-driven, no owned tables in v1)
Reporting/Moderation	User reports against profiles or messages, resolution tracking	Report
Administration & Audit	User/role/subscription management via User.Role; login/activity logs	(reads across modules) + LoginLog

4. Entity Catalog

Format: Purpose → Key Attributes → Primary/Foreign Keys → Business Rules → Security Classification. Entities are grouped by owning module.

Entity	User
Purpose	A person (or staff member) who can access the system.
Key Attributes	UserId, Email, MobileNumber, PasswordHash, Role, Status, EmailVerified, MobileVerified, FailedLoginAttempts, LastLogin, CreatedAt, UpdatedAt
Primary Key	UserId
Foreign Keys	—
Business Rules	Email and MobileNumber unique. Password stored only as a secure hash. Role ∈ {Candidate, Admin, Moderator}; account cannot log in unless Status = Active.
Security Class.	HIGH
Change from draft	<i>Added Role, replacing a separate Admin entity that was never defined in the draft.</i>

Entity	Session
Purpose	An active authenticated session/token.
Key Attributes	SessionId, UserId (FK), Token, ExpiresAt
Primary Key	SessionId
Foreign Keys	UserId → User
Business Rules	Expired sessions are rejected at the API gateway, not just the UI.
Security Class.	HIGH
Change from draft	<i>Newly catalogued — existed only in the ER image, undocumented in the draft.</i>

Entity	OTP
Purpose	One-time codes for phone/email verification and login.
Key Attributes	OtpId, UserId (FK), OtpCode, ExpiresAt, IsUsed
Primary Key	OtpId
Foreign Keys	UserId → User
Business Rules	An OTP can be consumed exactly once; expired or used codes are rejected.
Security Class.	HIGH
Change from draft	<i>Newly catalogued.</i>

Entity	LoginLog
Purpose	Audit trail of login attempts, for security review and the Administration/Audit module.
Key Attributes	LoginLogId, UserId (FK), LoginTime, IpAddress, LoginStatus

Primary Key	LoginLogId
Foreign Keys	UserId → User
Business Rules	Append-only; never updated or deleted by application code.
Security Class.	MEDIUM
Change from draft	<i>Newly catalogued.</i>

Entity	Profile
Purpose	Core, searchable matrimonial profile — trimmed to fields needed for search/matching; detailed sections split into dedicated entities below.
Key Attributes	ProfileId, UserId (FK), FirstName, LastName, Gender, DateOfBirth, MaritalStatus, Sect, SubSect, IsRevert, Education, Income, Country, State, City, HeightCm, Bio, VisibilityStatus, CompletionPercentage, CreatedAt, UpdatedAt
Primary Key	ProfileId
Foreign Keys	UserId → User (1:1)
Business Rules	One User owns one Profile. Must reach CompletionPercentage ≥ 60% before appearing in search. Female profiles default to private visibility. IsRevert feeds the men's subscription revert-discount rule; DateOfBirth and MaritalStatus feed the age/divorced/widowed discount rules (highest single discount only, no stacking). Country/State/City are a denormalized copy of the Current-type Address row, refreshed whenever that Address changes — kept here only so Search can filter by location without a join; Address is the source of truth.
Security Class.	HIGH
Change from draft	<i>Trimmed from the draft's 19 generic fields; SubSect and IsRevert added (both required by FORM_SPEC.md but missing from the draft); religious/lifestyle/marriage-intention/background fields moved to their own entities below. NativePlace removed — superseded by Address(AddressType=Native).</i>

Entity	ReligiousPreference
Purpose	All of a candidate's own religious-practice and lifestyle answers (Part 3 of the profile form) — kept separate from core Profile since it's rarely queried for search filtering and changes on a different cadence.
Key Attributes	ReligiousPreferenceId, ProfileId (FK, 1:1), Madhhab, SalahFrequency, ReligiosityLevel, HijabStatus, BeardStatus, Hifz, QuranReadingHabit, IslamicEducationLevel, StudyIslamInterest, HalalDiet, Fasting, Zakat, Smoking, Alcohol, HalalEntertainment, MixedGenderComfort, MusicViews, Nafaqah, WifeWorkingAttitude, MahramGuardian, HousingStatus, MahrCapacity, CareerAfterMarriage, ProposalContactPref, MahrExpectation, WeddingTypePreference, IslamicParentingStyle, RoleVision, DeenNotes
Primary Key	ReligiousPreferenceId
Foreign Keys	ProfileId → Profile
Business Rules	HijabStatus shown only if Gender=Female; BeardStatus only if Male. Nafaqah/WifeWorkingAttitude/MahramGuardian/HousingStatus/MahrCapacity shown only for Male profiles; CareerAfterMarriage/ProposalContactPref/MahrExpectation shown only for Female profiles. Smoking and Alcohol are the only two fields in this entity that are mandatory.
Security Class.	MEDIUM

Change from draft	<i>Entirely new — closes the Sprint 1 gap where ~30 required domain fields had no home in the draft schema.</i>
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Entity	ProfileLifestyleGoal
Purpose	Multi-select Islamic lifestyle goals.
Key Attributes	Id, ReligiousPreferenceId (FK), Goal
Primary Key	Id
Foreign Keys	ReligiousPreferenceId → ReligiousPreference
Business Rules	One row per selected goal.
Security Class.	LOW
Change from draft	<i>New — multi-select fields cannot be flat columns.</i>

Entity	ProfileWeddingRitual
Purpose	Multi-select wedding rituals/functions.
Key Attributes	Id, ReligiousPreferenceId (FK), Ritual
Primary Key	Id
Foreign Keys	ReligiousPreferenceId → ReligiousPreference
Business Rules	Populated only when WeddingTypePreference = Large wedding.
Security Class.	LOW
Change from draft	<i>New.</i>

Entity	MarriageIntention
Purpose	Timeline and household-formation intentions (Part 1, Marriage Intentions).
Key Attributes	MarriageIntentionId, ProfileId (FK, 1:1), WhenToMarry, OpenToRelocation, WantChildren, FamilyStructure, WhosSearching
Primary Key	MarriageIntentionId
Foreign Keys	ProfileId → Profile
Business Rules	WhenToMarry and WhosSearching are mandatory; the rest optional.
Security Class.	LOW
Change from draft	<i>New.</i>

Entity	ProfileChildrenDetail
Purpose	Children-related detail, only relevant to divorced/widowed candidates.
Key Attributes	Id, ProfileId (FK, 1:1 conditional), DivorceCertificate, ChildrenCount, ChildrenCustody, ChildrenDetailsText
Primary Key	Id

Foreign Keys	ProfileId → Profile
Business Rules	Row exists only if MaritalStatus ∈ {Divorced, Widowed}. DivorceCertificate required if MaritalStatus = Divorced.
Security Class.	HIGH
Change from draft	<i>New — the draft had no representation of this Sprint 1 field group at all.</i>

Entity	ProfileDisabilityType
Purpose	Multi-select disability/health-condition types.
Key Attributes	Id, ProfileId (FK), DisabilityType
Primary Key	Id
Foreign Keys	ProfileId → Profile
Business Rules	Populated only if the candidate answers Yes to the disability question.
Security Class.	HIGH
Change from draft	<i>New.</i>

Entity	ProfileLanguage
Purpose	Multi-select languages spoken.
Key Attributes	Id, ProfileId (FK), Language
Primary Key	Id
Foreign Keys	ProfileId → Profile
Business Rules	One row per language.
Security Class.	LOW
Change from draft	<i>New.</i>

Entity	ProfileHobby
Purpose	Multi-select hobbies/interests.
Key Attributes	Id, ProfileId (FK), Hobby
Primary Key	Id
Foreign Keys	ProfileId → Profile
Business Rules	One row per hobby.
Security Class.	LOW
Change from draft	<i>New.</i>

Entity	Education
Purpose	Education history entries.

Key Attributes	EducationId, ProfileId (FK), Degree, Institution
Primary Key	EducationId
Foreign Keys	ProfileId → Profile
Business Rules	Zero or more per profile.
Security Class.	LOW
Change from draft	<i>Newly catalogued — existed only in the ER image.</i>

Entity	Occupation
Purpose	Occupation history entries.
Key Attributes	OccupationId, ProfileId (FK), JobTitle, Company
Primary Key	OccupationId
Foreign Keys	ProfileId → Profile
Business Rules	Zero or more per profile.
Security Class.	LOW
Change from draft	<i>Newly catalogued.</i>

Entity	Address
Purpose	A profile's two address records: native place (family/ancestral origin) and current residence.
Key Attributes	AddressId, ProfileId (FK), AddressType (Native Current), City, State, Country
Primary Key	AddressId
Foreign Keys	ProfileId → Profile
Business Rules	Exactly two rows per profile — one AddressType=Native (primary) and one AddressType=Current (secondary). Values may be identical if the person has never relocated. Current is the row that gets denormalized onto Profile.Country/State/City for search.
Security Class.	MEDIUM
Change from draft	<i>Resolved (previously Open Issue #2): AddressType added to disambiguate from Profile's own location fields, rather than leaving the two overlapping and undocumented.</i>

Entity	PartnerPreference
Purpose	What a candidate is looking for in a partner (full Part 2 field set).
Key Attributes	PreferenceId, ProfileId (FK, 1:1), PreferredGender (locked), AgeMin, AgeMax, HeightMinCm, HeightMaxCm, PreferredSect, PreferredSubSect, PreferredMadhhab, PreferredSalah, PreferredReligiosity, HijabPref, BeardPref, PreferredMaritalStatus, AcceptChildren, PreferredProfession, EmploymentPref, PreferredEducation, OpenToAnyRegion, PreferredCountry, PreferredCity, PartnerShouldRelocate, LivingArrangement, ChildrenPref, PreferredTimeline, MotherTonguePref, WaliInvolvementExpected, OtherExpectations
Primary Key	PreferenceId
Foreign Keys	ProfileId → Profile

Business Rules	PreferredGender auto-set to the opposite of Profile.Gender and locked — not manually editable. AgeMin/AgeMax, PreferredSect and LivingArrangement mandatory.
Security Class.	LOW
Change from draft	<i>Primary key typo ("preferpartanerId") corrected to PreferenceId; expanded from 8 to the full ~20-field set from FORM_SPEC.md Part 2.</i>

Entity	Guardian
Purpose	A Wali — any relative (father, uncle, brother, appointed guardian) associated with a candidate, whether they hold their own account or are simply listed for display.
Key Attributes	GuardianId, ProfileId (FK, 1:1), UserId (FK, nullable), Name, Relation, Phone, ShowContactOnProfile, TrustLevel, Status, CreatedAt
Primary Key	GuardianId
Foreign Keys	ProfileId → Profile; UserId → User (nullable)
Business Rules	UserId is null when the Wali has not registered an account — that record is marked TrustLevel = DisplayOnly rather than Registered. Phone is verification-only unless ShowContactOnProfile = true.
Security Class.	HIGH
Change from draft	<i>Corrected from being accidentally documented under an entity called "MatchCandidate" in the draft. UserId made nullable; TrustLevel added to distinguish registered vs. display-only Walis, per Sprint 1's flagged gap.</i>

Entity	GuardianApproval
Purpose	Append-only log of a Wali's approve/revoke decisions on a specific conversation.
Key Attributes	ApprovalId, ConversationId (FK), GuardianId (FK), Side, Status, ReviewedAt, Notes
Primary Key	ApprovalId
Foreign Keys	ConversationId → Conversation; GuardianId → Guardian
Business Rules	Never updated in place — a new row is inserted for every decision, preserving full history. Both the initiator's and the receiver's guardian must independently be in Status = Approved for Conversation.CurrentApprovalStatus to allow messaging; either side revoking blocks new messages immediately without deleting history.
Security Class.	HIGH
Change from draft	<i>New entity, replacing Conversation's flat GuardianApprovalStatus column, which could not represent revoke-then-reapprove history or dual-side gating.</i>

Entity	VerificationRequest
Purpose	Verification submissions and trust-score history.
Key Attributes	VerificationRequestId, ProfileId (FK), VerificationType, Status, SubmittedAt, ReviewedAt, ReviewedByUserId (FK), Remarks, TrustScore
Primary Key	VerificationRequestId
Foreign Keys	ProfileId → Profile; ReviewedByUserId → User
Business Rules	Multiple verification attempts are allowed. TrustScore updates only after a successful review.
Security Class.	HIGH

Change from draft	<i>ReviewedByUserId now points at User (Role=Admin), closing the previously-undefined "ReviewerId" reference.</i>
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Entity	MatchCandidate
Purpose	A directional match: candidates surfaced for a given profile by running its PartnerPreference against the pool.
Key Attributes	MatchId, ProfileId (FK), MatchedProfileId (FK → Profile), CompatibilityScore, TrustScoreAtMatch, BlendedScore, MatchedProfileVerified, Status, GeneratedAt, UpdatedAt
Primary Key	MatchId
Foreign Keys	ProfileId → Profile; MatchedProfileId → Profile
Business Rules	Matching is directional — A's list and B's list are independent rows, potentially with different scores. BlendedScore combines CompatibilityScore with the matched profile's current TrustScore; when VerificationRequest.TrustScore changes, existing rows are updated in place rather than requiring regeneration. Unverified profiles still appear, flagged via MatchedProfileVerified = false.
Security Class.	MEDIUM
Change from draft	<i>Rebuilt from scratch (the draft's entity #9 slot held Guardian data by mistake). Score model resolved into three explicit fields instead of one ambiguous "Score" column.</i>

Entity	SubscriptionPlan
Purpose	Defines available subscription tiers.
Key Attributes	PlanId, PlanName, Price, Duration
Primary Key	PlanId
Foreign Keys	—
Business Rules	Price changes do not retroactively affect already-active subscriptions.
Security Class.	LOW
Change from draft	<i>Newly catalogued.</i>

Entity	Subscription
Purpose	A profile's subscription history.
Key Attributes	SubscriptionId, ProfileId (FK), PlanId (FK), Status, StartDate, EndDate, AutoRenew, CreatedAt
Primary Key	SubscriptionId
Foreign Keys	ProfileId → Profile; PlanId → SubscriptionPlan
Business Rules	Only one active subscription per profile at a time. Men's pricing applies the highest single eligible discount (age >55, divorced, widowed, or revert) — discounts never stack; women's accounts are free regardless of these factors.
Security Class.	MEDIUM
Change from draft	<i>No structural change; discount business rule made explicit here rather than living only in prose elsewhere.</i>

Entity	Payment
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Purpose	Individual payment transactions against a subscription.
Key Attributes	PaymentId, SubscriptionId (FK), Gateway, GatewayTransactionId, Amount, Currency, Status, PaidAt
Primary Key	PaymentId
Foreign Keys	SubscriptionId → Subscription
Business Rules	GatewayTransactionId must be unique.
Security Class.	MEDIUM
Change from draft	<i>No structural change.</i>

Entity	Conversation
Purpose	A messaging thread between two profiles.
Key Attributes	ConversationId, InitiatorProfileId (FK), ReceiverProfileId (FK), CurrentApprovalStatus, Status, CreatedAt
Primary Key	ConversationId
Foreign Keys	InitiatorProfileId → Profile; ReceiverProfileId → Profile
Business Rules	Starts in PendingApproval. Only opens to messaging once both sides' GuardianApproval history resolves to Approved. Blocked users cannot create new conversations.
Security Class.	HIGH
Change from draft	<i>GuardianApprovalStatus replaced by the denormalized CurrentApprovalStatus, sourced from the new GuardianApproval log.</i>

Entity	Message
Purpose	Individual chat messages within a conversation.
Key Attributes	MessageId, ConversationId (FK), SenderProfileId (FK), Content, MessageType, SentAt, ReadAt, ModerationFlag
Primary Key	MessageId
Foreign Keys	ConversationId → Conversation; SenderProfileId → Profile
Business Rules	Messages cannot be edited after sending. ModerationFlag is set by automated content moderation — independent of, and not a substitute for, guardian approval.
Security Class.	HIGH
Change from draft	<i>Removed the duplicate ApprovalStatus column; ModerationFlag added to separate AI content moderation from guardian oversight, two genuinely different mechanisms the draft had conflated.</i>

Entity	Attachment
Purpose	File attachments on a message.
Key Attributes	AttachmentId, MessageId (FK), StorageKey, FileName
Primary Key	AttachmentId
Foreign Keys	MessageId → Message
Business Rules	Accessed only via signed URLs, never a public path.

Security Class.	HIGH
Change from draft	<i>No structural change.</i>

Entity	Photo
Purpose	Profile photos.
Key Attributes	PhotoId, ProfileId (FK), StorageKey, ThumbnailKey, Visibility, ModerationStatus, UploadedAt
Primary Key	PhotoId
Foreign Keys	ProfileId → Profile
Business Rules	Stored privately; accessible only through signed URLs. Requires moderation approval before becoming visible to other users — visibility hiding must be enforced server-side, not just in the UI.
Security Class.	HIGH
Change from draft	<i>No structural change.</i>

Entity	PrivacySetting
Purpose	Per-profile visibility controls.
Key Attributes	PrivacySettingId, ProfileId (FK, 1:1), PhotoVisibility, ContactVisibility, ProfileVisibility
Primary Key	PrivacySettingId
Foreign Keys	ProfileId → Profile
Business Rules	Defaults are more restrictive for female profiles, consistent with Profile's default-private business rule.
Security Class.	HIGH
Change from draft	<i>Newly catalogued — existed only in the ER image.</i>

Entity	Report
Purpose	A user-filed report against either a profile or a message.
Key Attributes	ReportId, ReporterProfileId (FK), ReportedProfileId (FK, nullable), ReportedMessageId (FK, nullable), Reason, Status, ReviewedByUserId (FK), ReviewedAt, Resolution
Primary Key	ReportId
Foreign Keys	ReporterProfileId → Profile; ReportedProfileId → Profile (nullable); ReportedMessageId → Message (nullable); ReviewedByUserId → User
Business Rules	Exactly one of ReportedProfileId / ReportedMessageId is populated per report. Reviewed only by users with Role ∈ {Admin, Moderator}.
Security Class.	MEDIUM
Change from draft	<i>Newly defined — referenced in the draft's Relationship Matrix but never actually specified.</i>

5. Attribute Classification (Security & Privacy)

Every field flagged Sensitive or Encrypt = Yes needs a concrete storage decision before implementation — in particular, Email and MobileNumber must remain queryable for login/uniqueness checks, so “Encrypt: Yes” alone is not a sufficient spec (see Open Issue #1).

Entity	Attribute	Mandatory	Sensitive	Encrypt	Mutable
User	Email	Yes	Yes	Yes (searchable—see Open Issue #1)	Yes
User	MobileNumber	Yes	Yes	Yes (searchable—see Open Issue #1)	Yes
User	PasswordHash	Yes	Critical	Hash	Yes
User	Role	Yes	Medium	No	Rarely
User	Status	Yes	No	No	Yes
User	LastLogin	No	No	No	Yes
Profile	FirstName / LastName	Yes	Yes	No	Yes
Profile	DateOfBirth	Yes	High	Yes	No
Profile	Gender	Yes	No	No	No
Profile	MaritalStatus	Yes	Medium	No	Yes
Profile	IsRevert	No	Medium	No	Rarely
Profile	Income	No	High	Yes	Yes
Profile	Bio	No	Low	No	Yes
ProfileChildrenDetail	ChildrenDetailsText	No	High	Yes	Yes
Guardian	Name / Phone	Conditional	High	Yes (Phone)	Yes
Address	AddressType / City / State / Country	Yes	Medium	No	Yes
VerificationRequest	TrustScore	Yes	Medium	No	Yes
VerificationRequest	Remarks	No	Medium	No	Yes
MatchCandidate	BlendedScore	Yes	Low	No	Yes
Payment	Amount	Yes	No	No	No
Payment	GatewayTransactionId	Yes	Medium	Yes	No
Message	Content	Yes	High	Yes	No
Message	ReadAt	No	No	No	Yes
Photo	StorageKey	Yes	High	No (access-controlled, not encrypted at rest by default)	No
Photo	Visibility	Yes	Medium	No	Yes
Report	Reason / Resolution	Conditional	Medium	No	Yes

6. Relationship Matrix

Parent	Child	Cardinality	Notes
User	Profile	1:1	
User	Session	1:N	
User	OTP	1:N	
User	LoginLog	1:N	
User	Guardian	1:N (optional)	Only populated when a Wali holds an account
User	VerificationRequest	1:N	as reviewer (ReviewedById)
User	Report	1:N	as reviewer (ReviewedById)
Profile	PrivacySetting	1:1	
Profile	MarriageIntention	1:1	
Profile	ReligiousPreference	1:1	
Profile	ProfileChildrenDetail	1:1 conditional	Divorced/Widowed only
Profile	ProfileDisabilityType	1:N conditional	
Profile	ProfileLanguage	1:N	
Profile	ProfileHobby	1:N	
Profile	Education	1:N	
Profile	Occupation	1:N	
Profile	Address	1:2 (exactly Native + Current)	Resolved — no longer overlaps with Profile's denormalized location fields
Profile	PartnerPreference	1:1	
Profile	Guardian	1:1	
Profile	Photo	1:N	
Profile	VerificationRequest	1:N	
Profile	Subscription	1:N	
Profile	MatchCandidate	1:N	as seeker (ProfileId)
Profile	MatchCandidate	1:N	as candidate (MatchedProfileId)
Profile	Conversation	1:N	as initiator
Profile	Conversation	1:N	as receiver
Profile	Message	1:N	as sender
Profile	Report	1:N	as reporter
Profile	Report	1:N	as reported target (nullable)
ReligiousPreference	ProfileLifestyleGoal	1:N	
ReligiousPreference	ProfileWeddingRitual	1:N conditional	Large wedding only
Guardian	GuardianApproval	1:N	
Conversation	Message	1:N	
Conversation	GuardianApproval	1:N	One row per approve/revoke decision, per side

Parent	Child	Cardinality	Notes
Message	Attachment	1:N	
Message	Report	0..1	Optional — a report may target a specific message
SubscriptionPlan	Subscription	1:N	
Subscription	Payment	1:N	

7. Entity-Relationship Diagram

The full ER diagram is on the following landscape page (30 entities across 12 modules, color-coded by owning module). It supersedes UMLDiagramV1_HLERdia.png — every correction from Section 1 is reflected: MatchCandidate is properly modeled, Guardian is de-duplicated, GuardianApproval replaces the flat approval column, and the new Profile-related entities from the Sprint 1 field audit are present.

Colour key: red = Auth, blue = Profile core, teal = Religious/lifestyle, orange = Profile detail & Guardian, purple = Partner Preference, green = Matching, yellow = Subscription/Billing, grey = Reporting.

8. Cross-Cutting Business Rules

- **Profile completeness gate:** a profile must reach $\geq 60\%$ completion (`CompletionPercentage`, stored and recomputed on save) before it is eligible to appear in Search or Matching.
- **Female privacy default:** female profiles default to private visibility (`PrivacySetting.ProfileVisibility`); more restrictive photo/contact defaults apply to female profiles.
- **Signed-URL-only media:** Photo and Attachment are never served from a public path — visibility and moderation gates are enforced at the API/serialization layer, not the UI, so a hidden field can never leak in a raw JSON payload.
- **Subscription discounts (men only):** age > 55 (55%), divorced (50%), widowed (50%), revert (50%). Only the single highest-eligible discount applies — discounts never stack. Women's accounts are free regardless of these factors.
- **Dual-guardian conversation gate:** a Conversation only allows messaging once both the initiator's and the receiver's Guardian have an Approved decision in `GuardianApproval`; either side revoking blocks new messages immediately, without deleting message history.
- **Directional, trust-blended matching:** MatchCandidate rows are generated per-profile ($A \rightarrow B$ and $B \rightarrow A$ are independent rows/scores). `BlendedScore` combines `PartnerPreference` compatibility with the matched profile's live `TrustScore`; a verification-status change updates existing rows rather than requiring regeneration.
- **Fields never collected** (unchanged from the original product decision): sexual history, exact home address, credit score/debt, social media credentials, political affiliation, forced medical/immigration disclosures, criminal history beyond what's legally required, photos of children, skin colour as a required field, IQ results, menstrual/reproductive medical details. No entity above should ever grow a column for any of these without a deliberate, separately-reviewed product decision.

9. Open Issues & Deferred Scope

Genuinely unresolved items — flagged rather than guessed at, per the review process for this document.

#	Issue	Why it matters	Needs
1	Email/MobileNumber marked “Encrypt: Yes” but must remain queryable for login and uniqueness checks.	Flat column encryption breaks equality lookups. Needs a specific mechanism (e.g. deterministic/HMAC-indexed encryption, or a separate hashed lookup column) before implementation.	A decision from you on which encryption/indexing approach to use, or an infra/security consult if you want a second opinion.
2	Quick-Match Wizard (from FORM_FUNCTIONALITY.md) status is pending — you've confirmed a decision is coming but haven't made the call yet.	If it's reintroduced as a real pre-registration flow, its localStorage-based answers will need a reconciliation path into Profile/PartnerPreference once a user actually registers.	Awaiting your decision — flagged here so it isn't silently dropped or silently assumed in-scope.